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CSA-0407

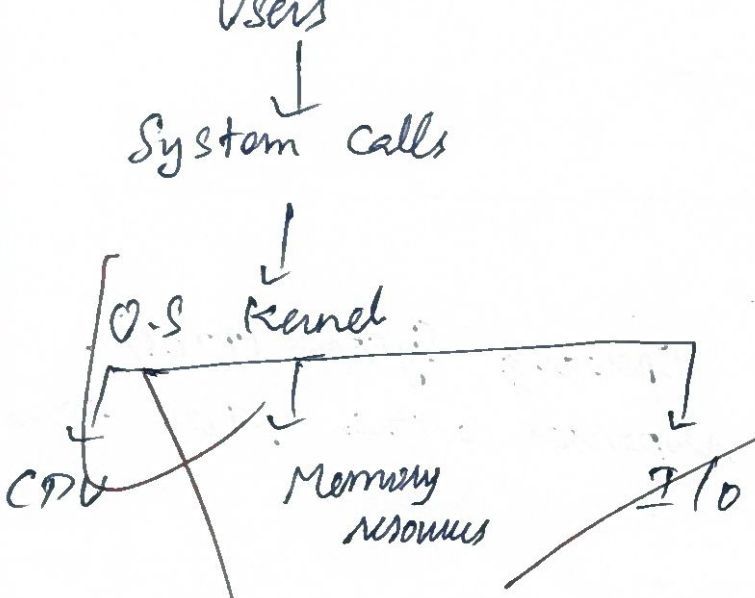
Class test — Operating Systems

1) a.) Types of O.S :-

Suitable OS types are Multi-user OS and Multi-tasking OS. They allow multiple users and processes to share CPU, memory and I/O resources efficiently.

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b.) Structure :-



c.) Process creation system calls :-

fork() → exec() → wait() → exit() //.

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2) a.) File and device system calls :-

It includes,

- File :- open(), read(), write(), close()
- Device :- ioctl(), read(), write().

b.) Layered structure

User programs
↓
System calls
↓
File system
↓
Device drivers
↓
Hardware

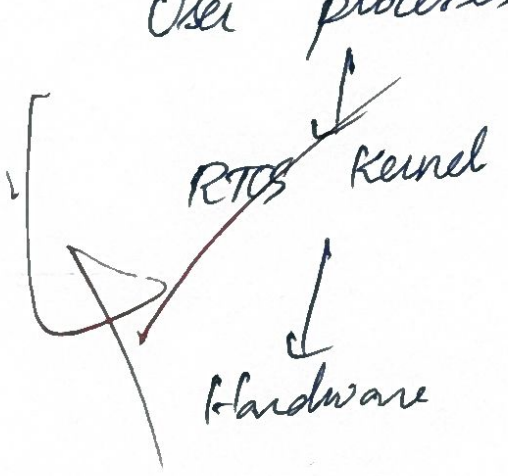
c.) File operation sequence

```
fd = open ("data.txt", O_RDONLY);
read (fd, buffer, 100);
close (fd); //
```

3) a.) RTOS:-

The suitable OS is Real-Time Operating System (RTOS). It provides fast and predictable responses within strict time limits.

b.) Structure

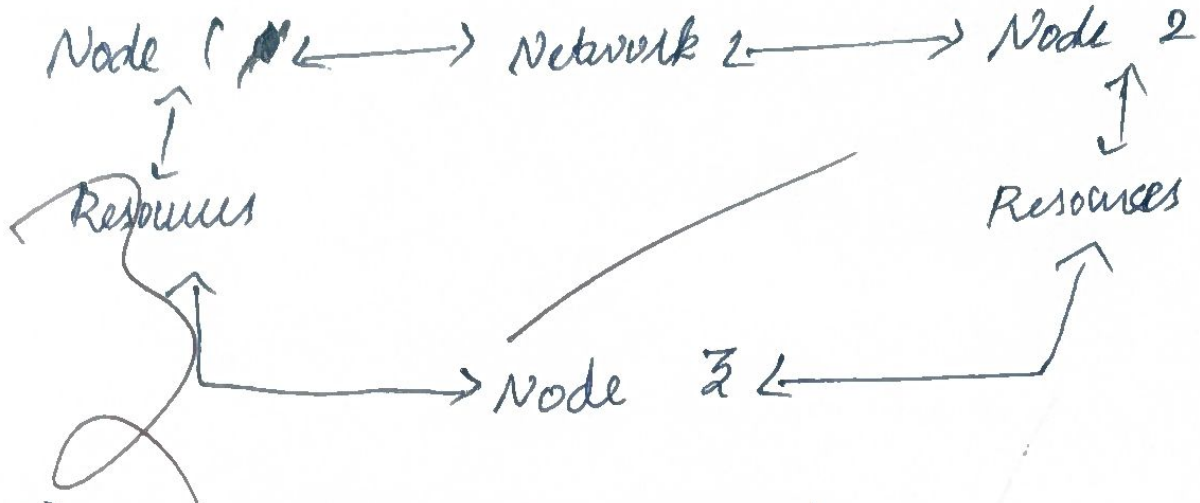


c.) Synchronization system functions:

- sem_wait() - waits for a semaphore.
- sem_post() - releases a semaphore
- pthread_mutex_lock() - locks a shared resource.

a.) A Distributed OS ... as a coordinated system. Unlike a centralized OS, resources and processing are distributed among multiple nodes.

b.) Structure:



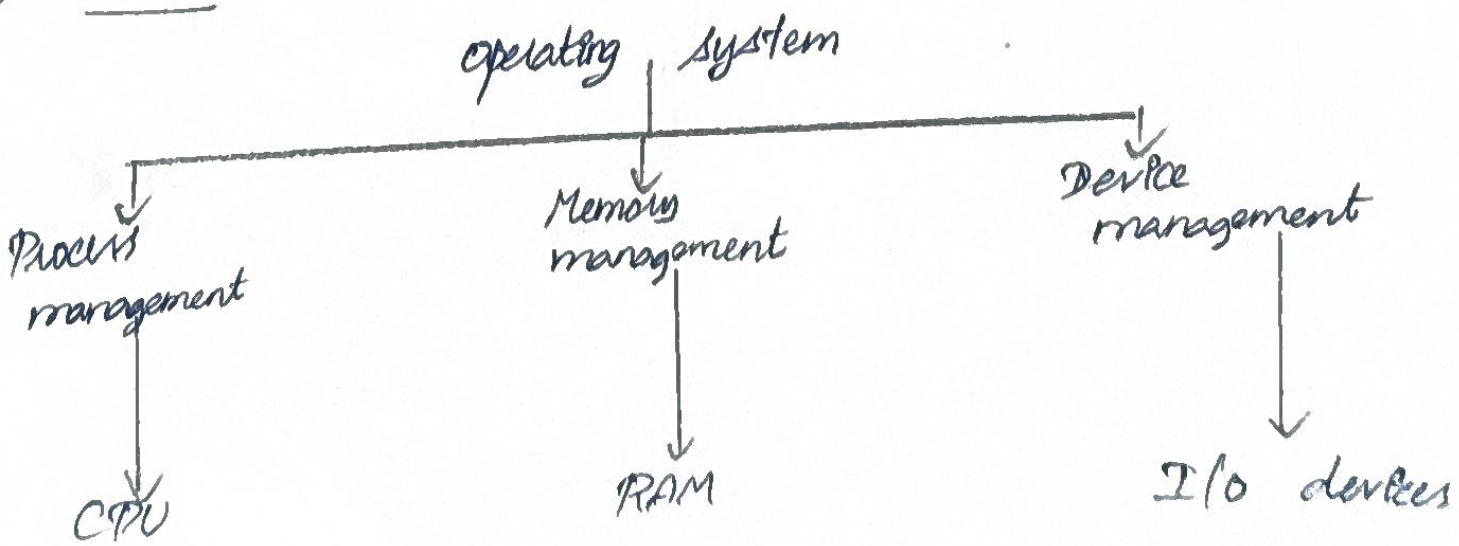
c.) IPC system-call sequence:

socket() → bind() → connect() → send() → receive() → close()//.

5.) a.) Major OS functions:

- Process scheduling
- Memory management
- Device management

b.) Structure:



(c)

malloc(100);

read (fd, buffer, 100);

write (fd, buffer, 100); //.